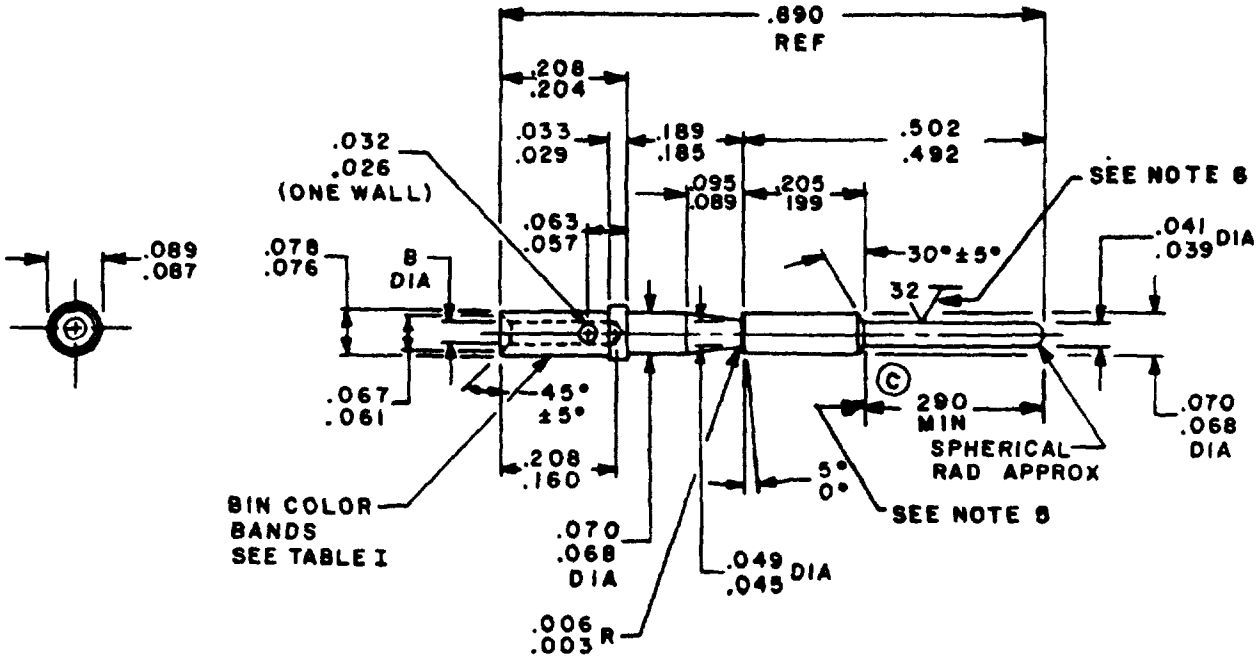
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE (FOR MIL-C-28840 CONNECTORS)

This specification sheet is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	0.08	.045	1.14	.076	1.93	.199	5.05
.006	0.15	.049	1.24	.078	1.98	.204	5.18
.026	0.66	.057	1.45	.087	2.21	.205	5.21
.029	0.74	.061	1.55	.089	2.26	.208	5.28
.032	0.81	.063	1.60	.095	2.41	.290	7.37
.033	0.84	.067	1.70	.160	4.06	.492	12.50
.039	0.99	.068	1.73	.185	4.70	.502	12.75
.041	1.04	.070	1.78	.189	4.80	.890	22.61

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Spherical end dimension to comply with MIL-C-39029.
5. Contact engagement area for localized finish option (see requirements).
© 6. Surface finish (electrical engagement area) 32 microinch finish. Pin tip flat as defined in MIL-C-39029 does not require 32 microinch finish.

FIGURE 1. Dimensions and configuration.

© denotes changes

FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, design characteristics, and configuration: See figure 1 and table I.

Tools: See table II.

© Localized finish option: Applicable contacts shall be furnished as specified in the localized finish requirements of MIL-C-39029. The E dimension of the Y area shall be .290 inch minimum. Nickel underplate required.

Vibration: Method 2005 of MIL-STD-1344, test condition III. Following the test, the contacts shall be tested for endurance in accordance with MIL-STD-167/1.

High impact shock: High impact shock test shall be conducted in accordance with MIL-C-28840, using a MIL-C-28840 qualified connector that has been approved for listing on the applicable qualified products list.

Mating contact: MIL-C-39029/84.

QPL evaluating activity: Naval Weapons Support Center, Crane, Indiana 47522.

© Porosity test localized finish: If localized finish is used, porosity shall be tested in accordance with method 1017 of MIL-STD-1344. Immersion in nitric acid is not applicable.

TABLE I. Design characteristics.

BIN code	B dia.	Color bands			Mating end size	Wire barrel size	Type	Class	C 1/ dia.
		1st	2nd	3rd					
508	.050 .048	Green	Black	Gray	20	20	A	B	.032 .026
450	.0355 .0335	Yellow	Green	Black	20	22	A	B	.032 .026
451	.0200 .0180	Yellow	Green	Brown	20	28	A	B	.022 .018

1/ Hole - one wall.

TABLE II. Tools.

BIN code	Basic crimping tool	Positioner	Installing tool	Removal tool
508	M22520/34-01	M22520/34-02	M81969/33-01	M81969/34-01
450	M22520/34-01	M22520/34-02	M81969/33-01	M81969/34-01
451	M22520/34-01	M22520/34-02	M81969/33-01	M81969/34-01

Military part number: Military part number shall be in accordance with table III.

TABLE III. Military part number.

BIN code	Military part number	Supersedes part number
508	M39029/83-508	
450	M39029/83-450	M39029/83-20-22
451	M39029/83-451	M39029/83-20-28

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

Low signal level contact resistance (type A): See table IV.

Contact resistance (type A) with silver plated wire: See table V.

Tensile strength (type A): See table VI.

TABLE IV. Low signal level contact resistance (type A).

Mating end size	Wire barrel size	Wire size	Maximum contact resistance (milliohms)	
			Initial	After conditioning
20	22	22 26	15 31	17 38
20	28	28 30	50 75	60 88

TABLE V. Contact resistance (type A) with silver plated wire.

Mating end size	Wire barrel size	Wire size range	Test current (amperes)	Maximum voltage drop (millivolts)					
				25°C ±3°C		25°C ±3°C After conditioning		200°C +3°C -0°C	
				Max	Max avg	Max	Max avg	Max	Max avg
20	22	22 26	5.0 2.0	73 53	66 48	88 64	75 55	125 91	94 69
20	28	28 30	1.5 1.0	54 60	49 55	65 73	56 62	92 103	69 78

TABLE VI. Tensile strength (type A).

Mating end size	Wire barrel size	Wire size	Axial load (pounds)				High strength copper alloy wire
			Silver tin-plated copper wires		Nickel-plated copper wire		
			Initial condition values	Thermal condition values	Initial condition values	Thermal condition values	
20	22	22 26	12 5	7.5 4.0	8 3	6.0 2.5	22 10
20	28	28 30	3 1.5	2.25 1.13	2 1.5	1.50 1.13	5.5 3.4

User activity:
Navy - OS

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5935-N264-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.